



2001 Network Tariff Report

Produced pursuant to clause 11.3.5 of Volume I of the Office of the Regulator General's Electricity Distribution Price Determination 2001 – 2005, and supplementary advisory notes

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1. Purpose

To provide guidance for customers, this Tariff Report sets out the issues, principles and rationale used by United Energy in setting its distribution tariffs for the year 2001. While extensive consultation has already taken place over the past year, comment is invited from those interested parties wishing to air their views in regard to future tariff reviews and on-going development of distribution initiatives. Over the coming five years, United Energy proposes to further develop its pricing initiatives, with particular emphasis on signals for encouraging the efficient and equitable use of the electricity distribution network while minimising environmental impact.

2. Executive summary

This report highlights the introduction of several innovations aimed at:

- providing residential & business customers with greater choice of tariffs;
- encouraging consumers to use energy more cost-effectively;
- reducing peak network demand and hence costs;
- substantially reducing fixed charges to all customers; and
- introducing an 'interruptible' load option for industrial customers

The introduction of these initiatives has been timed to coincide with the implementation of tariffs for 2001 required by the Office of the Regulator General.

For small to medium customers, UE has introduced a new LVkWToU (Smart meter¹) tariff. It utilises a type of meter that measures load every half-hour. Customers have an incentive to minimise usage during 3pm to 6pm summer workdays, by incurring a special charge for demand usage during that time period. The charge is called the Summer Demand Incentive Charge (SDIC). This new tariff has no daily fixed standing charge.

¹ Smart meters are also known as 'interval meters', which is the term used in the balance of this report.

United Energy has responded to calls to reduce fixed daily standing charges. All customers supplied by UE will have substantially reduced fixed distribution charges. For customers using an interval meter, fixed distribution charges have been completely abolished.

Also for these customers, the peak/off-peak time zones have been re-aligned increasing the number of hours per year that are classified as off-peak. The old off-peak periods are based on time zones that were set many years ago and no longer reflect current demand patterns. However, for those without interval meters, it was not practicable to change the time zones.

Seasonal components have also been incorporated into the tariffs. These, together with the changes to time-of-day zones, provide an incentive for customers to defer some of their consumption away from times where costly peak demands are most likely to occur.

Similarly, an interruptible tariff has been introduced for large customers who are willing to occasionally have some of their load remotely switched off, or stand-by generation started, by United Energy. This will enable more network capacity to be available when it is otherwise strained, by encouraging large customers to delay their usage by a few hours, in doing so, freeing-up capacity for others.

These initiatives will encourage alternatives to traditional network augmentation for meeting additional demand requirements.

The principles used by United Energy in the tariff-setting process are described in greater detail in sections of this report. They can be grouped under the following headings:

- compliance with the requirements of the Office of the Regulator General;
- greater customer choice and flexibility;
- market equity for all customers;
- reflecting the cost of supplying the network capacity
- encouraging some behavioural elasticity in the market;
- recognising the need for practicality; and
- seeking the most environmentally positive responses

Understandably, not all principles could be satisfied to their full extent, while fully complying with the ORG's requirements. UE however has sought to achieve as much as possible in satisfying these principles within the given constraints.

UE supports the ORG's pragmatic approach to tariff-setting – particularly recognising that there is no 'one best answer' or method. Rather, the ORG has set guidelines for determining an envelope of acceptability using upper and lower bounds, within which tariffs must fall.

Tariff-setting can be characterised as a somewhat 'nebulous' process. Tariffs for shared network services seek to equitably share the costs of owning, operating and maintaining the distribution network across a wide variety of customer configurations and usage patterns. It would be no surprise if UE's 2001 tariffs differ somewhat from those of other businesses. Each will have varying conditions to cater for, different methodologies and differing propensities for innovation.

3. Background

During the period from 1995 to 2000, network price signals have only been visible to customers consuming more than 160 MWh per annum. UE introduced several new tariffs improving signalling for the larger customers. Specifically the introduction of kVA demand-based tariff in 1998 has resulted in tangible improvements in network characteristics. Recent advances in metering technology enable much improved price signalling. This theme has been further developed in the tariffs applying from 1 January 2001.

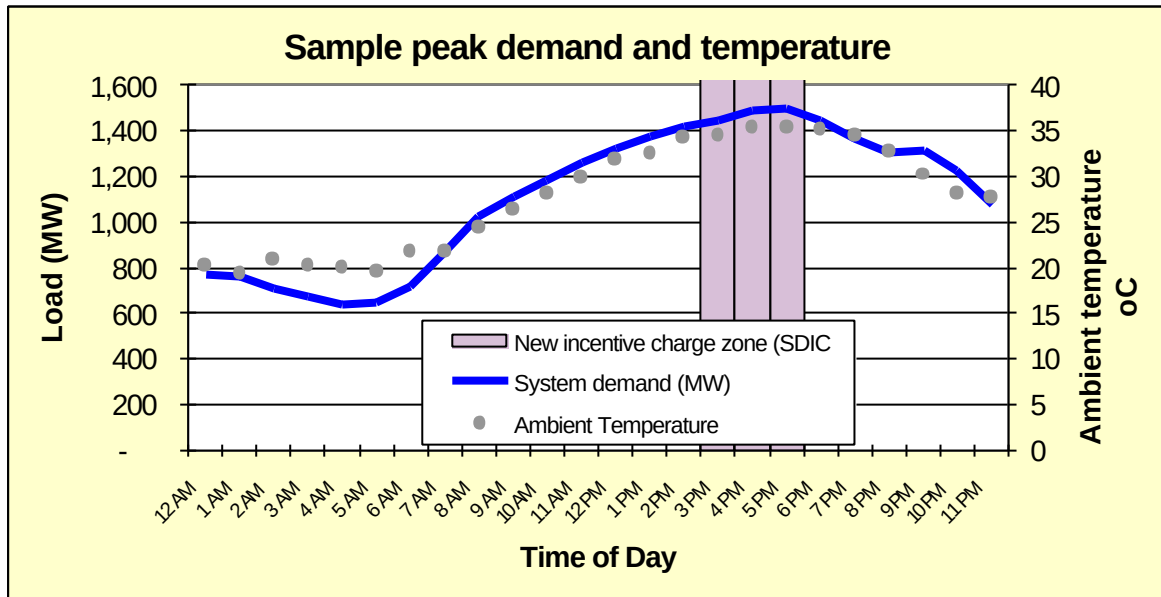
The review of distribution prices undertaken in 2000 has presented an opportunity to materially develop innovative responses to trends that have emerged in recent years. Some of the most significant trends include:

- the system maximum demand now occurring in Summer rather than Winter;
- the advent of retail contestability;
- increased focus on the environment effects of energy usage;
- advances in data-driven technology; and
- changes in the regulatory framework.

In the design of tariffs for the 2001, UE has aimed to incorporate the implications and opportunities that the trends afford.

Summer-peaking. The following chart is an illustration of the demand on UE's system during a recent peak Summer day

Chart 1 UE System load profile on a typical peak day



The chart illustrates that UE's demand is highly correlated with ambient temperature, and also that the peak coincides with the newly introduced 'Summer Demand Incentive Charge' (SDIC) time zone, aimed at reducing demand during this period.

Retail contestability. Over recent years, segments of the electricity market have been opened up to competition for energy supply. This commenced with larger customers, and is progressively making its way down to small customers. While not yet completed, the process of exposing customers to competition has also necessitated an effective separation in the industry of distribution, deemed to be non-contestable and therefore regulated, from retail energy marketing. This has also led to the requirement for increased 'transparency' in regard to the setting and application of distribution tariffs.

Environmental emphasis. There is now an increased interest on the part of many stakeholders in the promotion of more efficient and cost-effective use of energy. This applies to existing electricity generation and to distribution, consumption efficiency and the

use of alternatives. A particular emphasis has been on ‘renewables’, consistent with reduction of CO₂ emissions from fossil-fuelled power generation. This may imply a trend towards increasing interest in embedded generation over time.

Data technology. UE has utilised the recent advances in interval metering technology in the process of introducing the new LVkWToU tariff. Now even at the smaller customer level, half-hourly consumption data can be cost-effectively collected. This enables much more effective and cost-reflective pricing signals to be given to customers and for them to respond to make meaningful savings.

For smaller customers, the majority of metering in use today is based upon measuring and summing, over a period of up to 90 days, kWh usage totals. In some cases this is extended, with the use of a timeclock, to two totals, one for peak usage and the other for off peak usage, centered on a time zone of 7am to 11pm. This type of metering enables only the simplest of pricing signals to be imparted, and necessitates assuming that all customers of a like usage have the same average profile. By contrast, interval metering, logs usage data every half-hour, enabling much improved signalling and customer reward for beneficial changes in usage patterns. These may in future be able to be connected to a customer’s computer for load management and meter reading.

Also, UE has made increasing use of data acquisition technology on its own network, enabling more sophisticated planning techniques to be used and for initiatives to be better targeted and more cost-effective.

Regulatory framework. The alteration of the form of price control applied by the Office of the Regulator General of Victoria (ORG) away from ‘revenue yield’ to the ‘tariff basket’ approach, combined with the imposition of a high X-factor², has presented an opportunity to address the trends to a greater extent than might otherwise have been possible. UE’s commitment to harnessing advances in network data capture technologies and modelling of load profiles in particular, has enabled a much improved understanding of the key economic drivers and consumer behaviour. At the retail level, the discontinuation of the ‘maximum uniform tariffs (MUTs) gives rise to the opportunity to lessen some of the ‘distortions’ that

² A factor imposed by regulation that forces prices down over time in a ‘CPI-X regime’. For more details, see <http://www.reggen.vic.gov.au>

their use resulted in over time. UE has sought to have regard to the effects on customers of changes to its distribution tariff.

4. UE's tariff-setting obligations and constraints

Electricity Distributors are responsible for setting the network tariff component of the end-use customer's retail tariff. This component is referred to as the Network Use of System charge (NUOS). NUOS itself is comprised of the sum of Distribution (DUOS) and Transmission (TUOS) components.

The 2001 rates for these are contained in Appendix 1.

In the setting of these rates, an extensive set of 'rules' specified by the ORG, must be complied with. These rules limit the extent to which tariffs can change, particularly any average price increases. Within these constraints, distributors can design further changes that permit innovation and achievement of other objectives. Describing these objectives and the rationale in developing them is to a large extent the purpose of the balance of this report.

In cases where individual customers could be readily identified, the figures have been either omitted or aggregated into the closest equivalent group.

Average load patterns have been assessed on a customer group basis. Customers with load patterns that are materially different to the average may experience effects which vary somewhat to those described as applying in general.

Distribution tariffs are generally 're-packaged' to end-use customers by electricity Retailers. Retailers may not necessarily reflect these tariffs through to end-use customers. If the 'repackaging' is substantial, they may however create 'basis risk' for themselves and therefore have a disincentive to do so. The following discussion assumes that the distribution pricing signals are passed through to customers substantively unaltered.

5. Principles for distribution tariff setting (basis for tariff policies)

UE has aimed to have due regard to all the main factors and stakeholders in producing a practical set of tariffs. However, not all objectives can be met to their full extent simultaneously.

The design of the tariffs has sought to incorporate the following policy principles:

- **Regulatory compliance.** Compliance with the regulatory requirements of the Office of the Regulator General Victoria (ORG), as specified in and interpreted from Volumes I & II of the Electricity Distribution Price Determination, and subsequent partial re-determination.
- **Customer choice.** Where feasible provide customers with meaningful choices of tariff option based upon behavioural response and shared benefits. Having regard also however to avoiding the creation of 'loopholes'.
- **Market equity.** Having regard to 2000 price levels and hence relative changes. Also, pricing should apply to all retailers in a neutral manner and not impede the viability of full retail contestability.
- **Cost reflectivity.** Seek appropriate mechanisms for achieving inter-customer group equity in the allocation of distribution cost-recovery requirements. Pricing to recognise cost-reflectivity, within the constraints imposed by interpretation of regulatory requirements, customer group averaging assumptions, cost allocation methodologies employed and existing conditions, such as existing metering. Have due regard to reasonable economic bounds on upper and lower limits, particularly as pertaining to efficient investment signalling and network usage.
- **Behavioural elasticity.** Seek to utilise rational consumer behavioural elasticity, both in terms of usage pattern responses to pricing signals and tariff switching.
- **Practicality.** Seek to simplify mechanisms balancing against economic functionality, also having regard to minimising transaction and pricing administration cost.

- **Environmental.** Within the limitations of the scope and context of electricity distribution pricing, have regard to opportunities to improve utilisation efficiency and accommodation of emerging energy technologies, particularly in respect of greenhouse gas emission reduction.

6. Main features

The main features of the 2001 tariffs compared to those which previously applied are:

- The introduction of a new 'LVkWToU' ('interval' meter) tariff for medium and small customers
- The introduction of 'interruptible' tariffs for large customers
- Continuation of kVA-based tariffs for large customers
- The closure (to new connections) of some obsolete tariffs, including:
 - LVS2R, LV Dedicated, LVM2R5D, LVM2R7D

7. Tariff Structure

Fixed (standing daily) charges. UE has sought to align tariff structures with the key economic drivers, but also respond to the wishes of customer groups as expressed by the Government and ORG, in specifically reducing the proportion of charges collected as 'daily fixed charges'. Consistent mainly with these considerations, the standing charge for smart meter tariffs has been removed, and substantially reduced for other tariffs. This will tend to have the effect of favouring customers with lower than average annual consumption. The use of interval meters enables a direct assessment of a customer's maximum demand, as compared to using the 'proxy' of average load pattern and energy usage to determine the average demand. Hence, given that one of the intended main components of the standing charge is the recovery of demand/capacity-related costs, the smart meter enables the standing

charge to be reduced and offset by the use of demand/capacity charges such as the Summer Demand Incentive Charge (SDIC). To the extent that in some tariffs, the reduced fixed charge is offset by increases in the kWh-related charges, UE now has an increased exposure to volume volatility, particularly driven by weather-related phenomena. Short-term network costs are not particularly related to ambient temperature as they are largely driven by investment in assets. The SDIC also introduces additional volatility, to the extent that customers respond or do not respond to the pricing signal during times of high network load, for example sequences of hot summer days with high overnight temperatures.

kWh ‘blocks’. UE has discontinued the use of ‘stepped’ energy blocks from its tariffs. The benefit of this has been to simplify the data administration and to bypass the inherent shortcoming as a pricing signal when the meters are infrequently read, eg. at intervals of up to 90 days. This also has the effect of lessening the distinction between large kWh volume and low volume consumers. This also eliminates the tendency of incrementally reducing price blocks being similar in effect to a fixed charge.

Seasonalisation. In recent years, UE’s system peak demand has moved from occurring in the Winter and is now in the Summer period. This is likely to be due to the increasing proportion of temperature-related load, such as air-conditioning, in the small to medium sectors and is therefore likely to continue into the future. Accordingly, for all tariffs, the year has been divided into two seasons, Summer (November to March inclusive), and non-Summer (April to October inclusive). This applies to both peak energy charges and the SDIC. The merits of implementing ‘shoulder season’ zones was also considered, but not implemented in the interim in the interests of simplicity.

The summer is also of particular emphasis because due to network equipment thermal ratings and less effective cooling rates in summer, the network has a lower useable capacity than during the winter period. This further supports the need for summer-related signals.

Demand-based charges. Most distribution costs are ultimately driven by providing and maintaining the capacity required to supply the maximum demand on the system. With the advent of interval metering suitable for use with smaller customers, it is now possible to more directly measure a customer’s demand, compared to the ‘averaging’ that was required with single or two rate metering. Thus a more direct link between consumer behaviour and

the cost of supplying them can be established and reflected in pricing structure. Accordingly, for the 'interval meter' tariffs, a 'Summer Demand Incentive Charge' (SDIC) has been introduced together with reductions in kWh-related charges and standing charges. This will empower customers to modify their usage patterns, if they wish, so as to avoid times of peak demand, and thus reduce their costs. The SDIC applies to medium and large customers as well as those small customers who select the LVkWTOU tariff. Larger customers incur a lower rate for the SDIC but have an offsetting additional '12-month rolling average' demand charge which covers a wider time zone than the SDIC, in order to discourage the creation of new peaks at other times due to the re-instatement of deferred load.

Time zones (Demand-based charges). The SDIC is focussed on the time of day where the network capacity is most likely to be reached. Evaluation of historical data indicates that during 2001 the key 'time window' is between 3pm and 6pm on Summer weekdays (see chart 1). Similarly, the base for calculation of the 12-month rolling average demand charge is now based upon a time window of 7am to 7pm weekdays. This was previously based on the maximum demand at any time. Because usage patterns change over time, these time zones will be reviewed for appropriateness on an annual basis.

Time zones (kWh-based charges). As discussed in 'seasonalisation', for all tariffs, the year is now divided into two seasons, Summer (November to March inclusive) and non-Summer. In the case of interval meter tariffs, the peak time zone is from 7am to 7pm (Daylight Standard Time) on Summer workdays. For other tariffs the times applicable in 2000 continue, generally 7am to 11pm (Eastern Standard Time) for multi-rate and 'any time' for single-rate tariffs, as limited by the existing metering technology.

Ideally, all tariff time zones would have been standardised. However, the inherent limitations of infrequently read simple metering and time switch hardware, has unavoidably led to the previously applicable zones being preserved where required to support an existing tariff. In the case of tariffs where 'interval meters' are installed, the time zones have been aligned to a shorter period. Such meters also enable readings to be adjusted to reflect daylight saving. Similarly to the re-structured demand charges, the more focussed time zones are designed to encourage and reward cost-reducing behavioural responses from consumers.

Metering. By introducing the interval meter tariff to small consumers, UE has sought to utilise the benefits available from advances in metering technology. Much more accurate data on individual load patterns can now be gathered, avoiding the need for customers to be ‘averaged’ with a group of other customers and thus lessening the potential for cross-subsidisation between efficient and inefficient usage patterns. The use of interval metering also avoids the need to use average quarterly energy (kWh) consumption to estimate a customer’s maximum demand, and therefore cost to supply. This increases pricing accuracy, lessening risk and associated supply cost, enabling savings to be shared with consumers. It is expected that over time, interval metering will be used more and more commonly, and may ultimately enable a wide range of ‘smart’ services to be provided for customers. These may include load management/monitoring by home PC, on-line billing and payment, real-time energy price signalling.

Self-generation. A tariff suitable for non-stand-by type embedded generators that generally ‘export’ from their site is being developed. It will have the objective of ensuring that embedded or distributed generators be appropriately rewarded for avoided transmission costs, and also bear their share of distribution costs with other users of the network. For those large customers that are able to reduce their net load by running on-site generation to supply a portion of their needs, the 7am to 7pm time zone discussed earlier will encourage them to operate the generation during this period, and perhaps discontinue generating during the 7pm to 7am and non-workdays when alternate energy supply tends to be cheaper.

8. Fixed, variable and demand charge level breakdown

As discussed earlier, in line with calls from stakeholders, fixed daily standing charges³ have all been either substantially reduced, or in the case of those with an interval meter, eliminated completely.

As an example the standing charge of the LVS1R, incorporating those on ‘dedicated’, has reduced by around 38%. This pertains to around 500,000 residential customers. The

³ In some other constituencies, these are also known as ‘service availability charge’

corresponding proportions of ‘(%**Standing**: %**Variable**: %**Demand**)’ components before and after are, **(16%:84%:0%)** in 2000, compared to **(10%:90%:0%)** in 2001. The customers anticipated as moving from LVS1R to the new LVkWToU tariff would result in an average ratio of **(0%:93%:7%)**. Combined with the overall reduction in average rate for LVS1R, it can be seen that the proportion is now more weighted towards variable. This will favour low usage customers most.

The introduction of new time zones for the interval meter tariff provided the opportunity to some re-balancing of ‘peak’ and ‘off peak’ rates. This was undertaken using estimated average load patterns by customer group, and the peak/off-peak proportion of those patterns in the new time zones.

9. Rationale for the introduction of new tariffs, or closure of existing ones

Introduction of the new ‘low voltage kilowatt time of use’ (LVkWToU) tariff.

Traditionally, low energy users have tended to be subjected to a high degree of ‘averaging’ with other broadly similar customers. This can lead to undesirable outcomes such as cross-subsidisation as well as forecasting and hence business risk. Increased risk generally means increased costs and therefore ultimately increased prices. Until now the small customer has had very little choice available and little incentive to beneficially alter their usage patterns. Advances in metering technology, particularly interval meters that are suitable for use in the residential sector, together with improved network data gathering, has enabled the opportunity to introduce pricing mechanisms that are more closely aligned with network costs drivers than has traditionally been feasible. To the varying extent that customers will modify their usage patterns in response to rational pricing signals, the introduction of the LVkWToU tariff should produce the following effects over time:

- Improved average network utilisation
- Improved customer satisfaction through the exercising of choice and receipt of benefit for effort made
- The introduction of the SDIC empowers customers to make savings by altering the time of use of their consumption, for example shifting usage from peak times on summer weekdays to off peak times

- Improved levels of inter and intra-customer group equity thereby lessening cross-subsidisation
- Collection of much better quality data enabling more accurate planning
- Deferral of capital network capacity augmentation tending to lessen the regulated asset base over time, reducing costs
- Increase in transparency with consequential network cost saving that can be shared with the customer groups giving rise to the savings
- Enabling of future 'smart services' for customers through communication and automation capability
- Provides a technology 'platform' for 'demand-side management' and differentiated added-value customer services from electricity retailers
- Customer response should assist with improving the economic and environmental efficiency of the production, transmission, distribution and usage of electricity. Furthermore the tariff may assist in reducing CO₂ emissions through:
 - Increased uptake of energy efficient demand-side technologies (because the tariff provides an improved basis for the consumer to gain the benefits)
 - Improved usage patterns (higher load & seasonality factors) leading to the more efficient usage of power station capacity

Introduction of the new 'interruptible' LVkVAToU(I) & HVkVAToU(I) tariffs.

UE would be able to enhance its network management options if some loads in key constraint locations could be called upon for occasional temporary interruption in the event of a likely shortfall in supply capacity. Such shortfalls may be caused by unexpectedly high demand or perhaps a scheduled or unscheduled reduction in available capacity, for example caused by a vehicle accident. An interruptible load tariff for large customers has been introduced for both low voltage and high voltage supplies. The applicability will be assessed with the customer on a case-by-case basis, and will require the installation of remote load control equipment.

It is also intended that this tariff be suitable for those customers that have stand-by generation capacity available for use when called upon by the UE control room.

- customers can make considerable savings in demand charges by making all or part of their load available for occasional direct interruption by the UE control room at times of actual or imminent shortfall in network capacity
- customers can also gain a benefit from making available any suitable stand-by generation they may have (not intended for full-time generators)
- should encourage 'demand-side' management, enabling improvements in economic efficiency of supply, such as deferral of capital expenditure improved load factor

Closure to new connections of the following tariffs

The following tariffs have been closed to new connections

- **LVS2R**
- **LV Dedicated**
- **LVM2R5D**
- **LVM2R7D**
- **LVL1R (already closed)**
- **LVL2R (already closed)**

The introduction of the LVkWToU tariff has rendered a number of previous tariffs unnecessary. The following factors were taken into account in arriving at this decision:

- some tariffs were utilised by only a few customers
- some metering configurations required to sustain a tariff are less cost-effective than the 'interval meter' alternative that has much improved functionality and future utility
- some tariffs would have been incompatible with other tariffs, generally the new tariff is able to deliver similar features without the need for a multiplicity of specialised tariffs
- reducing the number of tariffs improves simplicity both from administration and customer awareness viewpoints
- the introduction of 'interval metering' enables customers to be distinguished according to their actual electrical load profile compared to the traditional approach of derived 'average profiles' by customer type

The purpose of closing some tariffs to new connections, rather than withdrawing them completely and migrating customers to the new ones, is to give time for the new tariffs and their features to be understood by customers. This means that customers with significantly poorer-than-average load factors, for example, will not experience ‘rate shock’. It is expected that over time, as customers become accustomed to them with the assistance of Retailers, increasing numbers will choose to move to the new tariffs. This will also have the effect of limiting the potential for customers with relatively poor usage patterns to experience high average price increases, as they are less likely to move away from their existing tariffs.

Continuation of kVA-based tariffs

The tariffs introduced in recent years for large customers, greater than 400 MWh/yr, have been retained and updated.

- Enables customers to make savings by improving their power factor
- Over time, this should reduce the amount of extra network capacity required
- More than 25% additional off-peak hours are now available
- SDIC to provide sharper signals and opportunity for behavioural elasticity
- Peak rolling demand was formerly ‘anytime’ and is now 7am –7pm workdays

The LVkWToU tariff for smaller customers is not kVA-based as there are relatively few practical methods by which smaller customers can respond to power factor signals. The meter cost is lower for kW measurement compared to the cost for measuring kVA.

10. Estimation of ‘upper’ & ‘lower’ bounds

As discussed by the ORG⁴, a myriad of alternative economic methodologies exist for estimating upper and lower bounds for network pricing and also for cost allocation to customer groups. Each methodology has its merits and shortcomings.

⁴ p198, Volume I, ‘Statement of Purpose and reasons’, *Electricity Distribution Price Determination 2001 to 2005*.

A major factor impacting on the selection and/or development of a methodology is the availability and quality of information. While UE has made considerable advances in recent years in respect of gathering greater levels of data about its network and key drivers such as ambient temperature, a number of assumptions, simplifications and assessments were required in order to estimate some limits for responding to the ORG's guidelines methodology. In UE's view, the methodology has limitations both in respect of mimicking anything that is likely to occur in real-life instances, and as to definition. It is agreed however that there is some utility in serving as a general indication of tariff 'reasonableness' as to average price by customer category.

Key aspects of the methodology used:

The following was the basis for assessing whether proposed tariffs were within the bounds specified by the ORG, as opposed to explicitly being directly used to set rates.

Load profile data. Average load patterns corresponding to the key customer groups were derived from historical data such as SCADA, zone sub, or terminal station interval data. Sections of network that are predominantly of a particular customer category, with enough customers to form a viable sample (eg. Residential) were selected. Three primary load patterns were identified:

- small (during periods of average climatic conditions)
- small (during periods of high ambient temperature sequences)⁵
- medium/large

These patterns were compared to the system profile, at the terminal station level, and the degree of contribution to overall peak assessed.

Patterns were also used to estimate other average parameters such as load factor, from which a relationship between metered energy, compensating for energy losses & power factor and 'coincident maximum demand' (CMD) can be derived, as well as 'anytime maximum

⁵ By comparing the 1st and 2nd patterns, and making an estimate of the penetration proportion of customers with temperature-related loads such as refrigerative air-conditioners, it is possible to derive a pattern for small customers with and without such appliances.

demand' (AMD). Diversity was also examined, within the limitations of feeder to feeder data.

These relationships are important in bridging the gap between data collection limitations and providing an appropriate basis for cost allocation.

In the case of public lighting unmetered supplies, a load pattern was assessed from installed equipment inventory, wattage ratings and daylight hours.

When combined with metered kWh data, various parameters by customer segment were quantified, for use as cost-allocators in the subsequent analysis.

Asset data. Various customer segments are connected at different levels of embedment in the network, relative to the terminal station interface with the transmission grid. Therefore some segments, such as unmetered lighting require the whole asset chain, as well as assets that only they use, whereas others, for example High Voltage supply, require only a much smaller part of the system to be in place in order to achieve supply to their premises⁶. When combined with information from the network asset database, an 'asset participation matrix' was derived, in units of regulated⁷ asset value. **Table 2** 'Asset participation matrix'

⁶ Subject to the proviso that it is often the case that a given customer benefits from being connected to a diversified, geographically comprehensive network, giving rise to an element of 'common good' shared benefit.

⁷ The ORG has specified a methodology (see www.reggen.vic.gov.au) for determining the depreciated asset base to which a rate of return is applied.

Tariff name	Asset level					
	SubTrans	High V	Low Voltage (LV)			Public lighting
	SubT	High V	Large	Medium	Small	Unmetered
LVS1R LVS2R LVDed	LV Small					
LVM1R LVM2R5D LVM2R7D	LV Medium					
LV Unmetered LV Unmetered PL LV Unmetered non PL Watchman lighting	LV Unmetered					
LVKWTOU (sml:med)	LVS & M on new smart meter tariff					
LVL2R.KW LVL2R LVL1R.KW LVL1R LVL2R-KVA LVKVATOU (lge) LVKVATOU Interruptible	LV Large					
HV.KW HV-KVA HV-KVA Interruptible	HV					
ST22.KW ST22.KVA	ST					

Shaded areas denote that a particular asset level is used by a particular tariff group.

Cost data. Historical and forecast cost data associated with owning, maintaining and operating the regulated distribution assets were compiled.

- **O&M (direct)**
- **Depreciation**
- **Transmission**
- **other (indirect)**

Cost allocation. Costs, other than transmission, were allocated on the basis of each group's usage of each level of network asset using the parameters outlined above. In the case of Transmission, similar methodology was used except that around 1/5th was allocated on the basis of kWh usage, which approximates the transmission charges levied more closely than using a demand basis alone.

11. The interpretation of the ORG's requirements in assessing 'upper & lower' bounds

The ORG has stated that its intention was not to be ‘prescriptive’ as to detailed methodology. Being mindful of this, and practical limitations in respect of data, UE has made the following assumptions.

- **Class of customer** means each tariff of like connection level and capacity, as opposed to individual tariffs.
- **Growth** in customer numbers, demands, kWh usage is assumed to be zero, therefore growth-related Capex can be excluded, other Capex being approximated by depreciation.
- The removal of an entire customer class is hypothetical in nature. In order to produce feasible numbers, it has been assumed that when a class is removed, there are no asset **stranding, decommissioning or reinstatement** costs. In reality these would be significant.
- In the case of ‘bypass’ assessment, **undepreciated** asset values have been used. This is because, physical bypass would generally necessitate the construction of new assets and therefore comparison with undepreciated distribution assets is more meaningful.
- The hypothetical removal of entire customer classes would also significantly affect demand **diversity** at various levels in the network. The current system load profile is a weighted average of customer groupings that have differing load usage patterns. Alteration of the weightings, is likely to significantly alter the system profile remaining, particularly relative contributions to peak capacity requirements, time of peak and asset utilisation. Accordingly, allowances were made for these interactions.
- **Asset participation** The analysis is sensitive to interactions between customer groups. If all LVS customers were removed, much of the LV network would still be required in order to provide geographically comprehensive supply to the unmetered public lighting class. The theoretical savings due to reduced kVA capacity requirement is likely to be not nearly as material as the need to preserve ‘geographic comprehensiveness’ to supply public lighting. Similarly, removing very small numbers of customers from a class is not likely to result in any savings. In order to proceed, UE has grouped sub-tariffs into main categories, but interpretation of the results should be mindful of these issues in drawing any conclusions.

Having applied the approach described, the results were as follows:

Table 3. Estimated ‘Avoidable’ and ‘Stand-alone’ cost of service

Tariff group	‘Avoidable’ (lower bound)	UE Tariff	‘Stand-alone’ (upper bound)
	Average effective cents/kWh NUOS excluding GST	Average effective cents/kWh NUOS excluding GST	Average effective cents/kWh NUOS excluding GST
LV Unmetered	1.7	3.0	22.0
LV Small ⁺	0.6	4.9	9.8
LV Medium ⁺	0.6	5.3	9.0
LV Large	0.5	3.2	4.9
HV	0.4	1.8	3.1

⁺ The new LVkWToU tariff has been apportioned into these tariff groups

In some instances, the rate representing the bulk of a group has been used

Given the limitations discussed earlier, it can be seen that the tariff groups are close to the mid-range and therefore comply with the ORG’s requirements of being between the two limits. Unmetered supplies may appear low compared to the upper bound, but this is due to diversity and geographic comprehensiveness issues described earlier, being a small group relative to other LV groups. Also, the LV Unmetered group incorporates assets that are not used by any other group⁸, resulting in a relatively high lower bound. In the presence of the other LV loads, particularly LVS, the unmetered group currently contributes little to the system peak at present and the appropriate tariff is therefore likely to be closer to the lower bound than the upper bound. This may however change in future if it becomes evident that the time of peak is trending towards coinciding with dusk.

12. Signals for efficient consumption

As mentioned earlier, a number of considerations, including economic efficiency, have been balanced against each other in the process of deriving tariffs. In UE’s view, the embracing of new technology, in particular ‘interval meters’, and the design of tariffs to utilise their potential to empower the demand-side to respond to pricing signals demonstrates a desire to improve consumption efficiency and efficiency of asset utilisation.

The overall downward pressure on distribution prices can be argued to reduce the incentive for consumers to improve the efficiency of consumption. UE has sought to mitigate this effect by providing more effectively structured tariffs, particularly targeting inefficient consumption at peak times. Having done this, there may be a lessened incentive to not ‘waste’ energy during non-peak periods. This same downward pressure may also have the effect of delaying the onset of energy alternatives.

Much of the consumer-behavioural debate will be affected by the price and structure that the final consumer sees, after it has been ‘repackaged’ by retailers, and so to that extent is beyond the direct control of distributors. The progressive introduction of full retail contestability may encourage greater levels of innovation amongst retailers as well as, arguably, an increased propensity to mitigate ‘basis’ risk by reflecting purchase cost structures through to customers to a greater extent than at present.

The introduction of interruptible tariffs may also encourage the market to respond in a more efficient manner to supply-side efficiency drivers.

13. Allocation of Transmission costs

To further encourage customer responses aimed at reducing peak demand and to ensure appropriate packaging of transmission pricing signals, all transmission tariffs have been set using only peak energy and demand components. Similarly the change in network time zones has necessitated some ‘re-balancing’ of rates to compensate for the charge being recovered over a smaller number of hours. Further re-balancing is possible in this area in the future.

Retail interest on this topic appears to be primarily with regard to alternative generation. It is appreciated that the presence of generators may produce some beneficial cost savings, particularly in the area of Transmission. It also appears likely that some distribution costs may even increase somewhat, particularly as some generators may necessitate additional

⁸ This can be seen in table 2

services such as backup, protection and power quality issues. There is also a need to consider equity between the 'pool' of all parties that share benefit from being connected to the diversified distribution system.

The 2001 pricing has not been developed with generation specifically in mind, other than in respect of encouraging those customers with stand-by generation to make it available for operation where this could be of material value during occasional events of prospective or actual network constraint. Alternative generation is currently extremely rare and from current indications as to viability is not expected to have a material bearing on UE's system during the pricing period covered by this report. UE's intention has been to be neutral between parties wishing to make use of its network, including embedded generators. Over the next year some focus will be given to developing a more detailed understanding of the issues involved and the further development of appropriate 'use of network' arrangements.

Nonetheless, UE considers that its introduction of interval metering to the smaller customer segment, may make the benefits to be gained from small size generators more identifiable and for the benefits to be shared with customers.

14. Changes in average annual charges to each segment

Table 4. Estimated change in effective tariffs (average annual charge/customer)

Tariff group	2000 Tariff		2001 Tariff		% difference
	Average annual cost excluding GST (\$/customer/year)	annual NUOS	Average annual cost excluding GST (\$/customer/year)	annual NUOS	
LVS1R	\$ 264		\$ 235		-10.8%
LVS2R*	\$ 278		\$ 278		0.0%
LV Ded*	\$ 48		\$ 47		-2.9%
LVM1R	\$ 929		\$ 799		-14.0%
LVM2R5D*	\$ 2,952		\$ 2,421		-18.0%
LVM2R7D*	\$ 2,557		\$ 1,855		-27.5%
LV Unmetered	**		**		0.2%
LV1R*	\$ 17,405		\$ 17,264		-0.8%
LV2R*	\$ 34,503		\$ 28,265		-18.1%
LVL2RkVA	\$ 46,197		\$ 45,752		-1.0%
HVkVA	\$ 209,604		\$ 206,042		-1.7%
STkVA	***		***		-2.1%
Average	\$ 473		\$ 423		-10.5%

The new LVkWToU tariff has been apportioned into the appropriate 'parent' tariff groups

* These tariffs are closed to new connections

** It is not meaningful to use customer numbers for unmetered supplies

*** This category has too few customers to show an average charge

Appendix 1

The distribution (DUOS) and transmission (TUOS) tariffs for 2001

Schedule of **Network** Use of System tariffs (**NUOS**) from 1.1.2001, excluding GST

Schedule of **Distribution** Use of System tariffs (**DUOS**) from 1.1.2001, excluding GST

Schedule of **Transmission** Use of System tariffs (**TUOS**) from 1.1.2001, excluding GST

NETWORK tariffs (Distribution + Transmission)

Schedule of Network Use of System (NUOS) Tariffs: 1 January 2001											
Network Tariff Component											
Rates Exclude GST	Standing Charge	Summer Demand Incentive Charge **	Rolling Peak Demand Charge	Summer Demand Incentive Interruptible Contract	Rolling Peak Demand Interruptible Contract	Summer Peak Energy Charge	Non-Summer Peak Energy Charge	Off Peak Energy Charge	Eligibility (consumption)		Minimum Chargeable Demand
		\$/kW pm or \$/kVA pm									
Tariff Name	\$/cust pa		\$/kVA pa	\$/kVA pm	\$/kVA pa	c/kWh	c/kWh	c/kWh	kVA	MWh	kVA
LVS1R	24.640	-	-	-	-	5.567	4.640	-	0-20	0-70	-
LVS2R*	35.510	-	-	-	-	7.736	6.445	1.356	-	-	-
LVDed*	-	-	-	-	-	-	-	1.356	-	-	-
LVM1R	32.032	-	-	-	-	7.640	6.367	-	20-150	70-400	-
LVL1R*	32.032	-	-	-	-	5.421	4.518	-	-	-	-
LVM2R5D*	46.163	-	-	-	-	9.294	7.746	1.356	-	-	-
LVL2R*	46.163	-	-	-	-	8.254	6.879	1.356	-	-	-
LVM2R7D*	46.163	-	-	-	-	9.014	7.512	1.356	-	-	-
LVUnmetered	-	-	-	-	-	7.553	6.295	1.356	-	-	-
LVkWTOU	-	8.873	-	-	-	8.353	6.962	1.803	0-150	0-400	0
LVkVATOU	-	4.684	53.402	-	-	1.959	1.632	0.895	>150	>400	150
LV Interruptible	-	4.684	53.402	(4.684)	(8.010)	1.959	1.632	0.895	>150	>400	150
HVkVATOU	-	3.174	36.180	-	-	1.384	1.154	0.583	>150	>400	1150
HV Interruptible	-	3.174	36.180	(3.174)	(5.428)	1.384	1.154	0.583	>150	>400	1150
SubTkVATOU	-	0.634	7.233	-	-	0.442	0.368	0.224	>150	>400	11.100

S = Small. M = Medium. L = Large. LV = Low Voltage. HV = High Voltage. SubT = Subtransmission. TOU = Time of Use. R = rate. D = Day

* indicates tariff is closed to new connections. New connections requiring off peak network charge rates must connect to one of the Time of

**Summer Demand Incentive Charge Transition Adjustment (Excluding GST)					
NUOS SDIC Adjustment (\$/kW/month or \$/kVA/month)					
Tariff Group	Jan-01	Feb-01	Mar-01	Nov-01	Dec-01
LVkWTOU	(8.873)	(8.873)	(8.873)	(4.437)	(4.437)
LVkVATOU	(4.684)	(4.684)	(4.684)	-	-
HV-KVA	(3.174)	(3.174)	(3.174)	(1.587)	(1.587)
ST22KVA	(0.634)	(0.634)	(0.634)	-	-

DISTRIBUTION tariffs

Schedule of Distribution Use of System (DUOS) Tariffs: 1 January 2001											
Distribution Tariff Component											
Rates Exclude GST	Standing Charge	Summer Demand Incentive Charge **	Rolling Peak Demand Charge	Summer Demand Incentive Interruptible Contract	Rolling Peak Demand Interruptible Contract	Summer Peak Energy Charge	Non-Summer Peak Energy Charge	Off Peak Energy Charge	Eligibility (consumption)		Minimum Chargeable Demand
Tariff Name	\$/cust pa	\$/kW pm or \$/kVA pm	\$/kVA pa	\$/kVA pm	\$/kVA pa	c/kWh	c/kWh	c/kWh	kVA	MWh	kVA
LVS1R	24.640	-	-	-	-	4.512	3.760	-	0-20	0-70	-
LVS2R*	35.510	-	-	-	-	6.680	5.566	1.356	-	-	-
LVDed*	-	-	-	-	-	-	-	1.356	-	-	-
LVM1R	32.032	-	-	-	-	6.497	5.414	-	20-150	70-400	-
LVL1R*	32.032	-	-	-	-	4.279	3.566	-	-	-	-
LVM2R5D*	46.163	-	-	-	-	8.151	6.793	1.356	-	-	-
LVL2R*	46.163	-	-	-	-	7.112	5.927	1.356	-	-	-
LVM2R7D*	46.163	-	-	-	-	7.871	6.559	1.356	-	-	-
LVUnmetered	-	-	-	-	-	6.411	5.343	1.356	-	-	-
LVkWTOU	-	7.570	-	-	-	7.211	6.009	1.803	0-150	0-400	0
LVkVATOU	-	4.684	38.446	-	-	1.074	0.895	0.895	>150	>400	150
LV Interruptible	-	4.684	38.446	(4.684)	(5.767)	1.074	0.895	0.895	>150	>400	150
HVkVATOU	-	3.174	20.927	-	-	0.699	0.583	0.583	>150	>400	1150
HV Interruptible	-	3.174	20.927	(3.174)	(3.140)	0.699	0.583	0.583	>150	>400	1150
SubTkVATOU	-	0.634	1.280	-	-	0.268	0.223	0.224	>150	>400	11,100

S = Small. M = Medium. L = Large. LV = Low Voltage. HV = High Voltage. SubT = Subtransmission. TOU = Time of Use. R = rate. D = Day

* indicates tariff is closed to new connections. New connections requiring off peak network charge rates must connect to one of the Time of

**Summer Demand Incentive Charge Transition Adjustment (Excluding GST)					
NUOS SDIC Adjustment (\$/kW/month or \$/kVA/month)					
Tariff Group	Jan-01	Feb-01	Mar-01	Nov-01	Dec-01
LVkWTOU	(6.852)	(6.852)	(6.852)	(3.426)	(3.426)
LVkVATOU	-	-	-	-	-
HV-KVA	-	-	-	-	-
ST22KVA	-	-	-	-	-

TRANSMISSION tariffs

Schedule of Transmission Use of System (TUOS) Tariffs: 1 January 2001											
Transmission Tariff Component											
Rates Exclude GST	Standing Charge	Summer Demand Incentive Charge **	Rolling Peak Demand Charge	Summer Demand Incentive Interruptible Contract	Rolling Peak Demand Interruptible Contract	Summer Peak Energy Charge	Non-Summer Peak Energy Charge	Off Peak Energy Charge	Eligibility (consumption)		Minimum Chargeable Demand
		\$/kW pm or \$/kVA									
Tariff Name	\$/cust pa	pm	\$/kVA pa	\$/kVA pm	\$/kVA pa	c/kWh	c/kWh	c/kWh	kVA	MWh	kVA
LVS1R	-	-	-	-	-	1.055	0.880	-	0-20	0-70	-
LVS2R*	-	-	-	-	-	1.056	0.879	-	-	-	-
LVDeD*	-	-	-	-	-	-	-	-	-	-	-
LVM1R	-	-	-	-	-	1.143	0.953	-	20-150	70-400	-
LVL1R*	-	-	-	-	-	1.142	0.952	-	-	-	-
LVM2R5D*	-	-	-	-	-	1.143	0.953	-	-	-	-
LVL2R*	-	-	-	-	-	1.142	0.952	-	-	-	-
LVM2R7D*	-	-	-	-	-	1.143	0.953	-	-	-	-
LVUnmetered	-	-	-	-	-	1.142	0.952	-	-	-	-
LVkWTOU	-	1.303	-	-	-	1.142	0.953	-	0-150	0-400	0
LVkVATOU	-	-	14.956	-	-	0.885	0.737	-	>150	>400	150
LV Interruptible	-	-	14.956	-	(2.243)	0.885	0.737	-	>150	>400	150
HVkVATOU	-	-	15.253	-	-	0.685	0.571	-	>150	>400	1150
HV Interruptible	-	-	15.253	-	(2.288)	0.685	0.571	-	>150	>400	1150
SubTkVATOU	-	-	5.953	-	-	0.174	0.145	-	>150	>400	11,100

S = Small. M = Medium. L = Large. LV = Low Voltage. HV = High Voltage. SubT = Subtransmission. TOU = Time of Use. R = rate. D = Day

* indicates tariff is closed to new connections. New connections requiring off peak network charge rates must connect to one of the Time of

**Summer Demand Incentive Charge Transition Adjustment (Excluding GST)					
NUOS SDIC Adjustment (\$/kW/month or \$/kVA/month)					
Tariff Group	Jan-01	Feb-01	Mar-01	Nov-01	Dec-01
LVkWTOU	(2.021)	(2.021)	(2.021)	(1.011)	(1.011)
LVkVATOU	(4.684)	(4.684)	(4.684)	-	-
HV-KVA	(3.174)	(3.174)	(3.174)	(1.587)	(1.587)
ST22KVA	(0.634)	(0.634)	(0.634)	-	-

Appendix 2

Terms used:

DSM	Demand-side management. Refers to the ability of customers that consume a product to change behaviours that alter the overall balance between supply and demand. Examples may include the installation of energy management systems, or deferring consumption at time of peak demand.
Load factor	Load factor is a measure of the extent to which average usage compares to maximum usage. It is generally much more efficient to supply a high load factor (100%=constant unchanging demand) than a 'peaky' (less than 100%) usage pattern. This is because depending upon the time of the peak, the capacity needs to be there all the time to supply the peak, even though most of the time it may not be fully required.
LVS1R etc	Abbreviated Tariff names. The 1 st two letters refer to the voltage level at which the customer is supplied, LV = low voltage 230v, HV = high voltage 6.6 or 11kV, ST = sub-transmission 22kV and above. The 2 nd section refers to the type of customer, S = small, M = medium, L = large, no letters generally indicates 'generic' interval metered customer. 3 ^d section indicates type of rate structure, 1R = single energy rate, 2R = peak/off-peak rate, kW = interval metered in kW, kVA = interval metered in kVA. Last section if present denotes interruptible (I)
kW	Kilowatts , (1,000 Watts) the ISO unit of power measurement
kVA, and power factor	kilo-Volt-Ampere , the vector product of voltage times current. It is a measure of the proportion of useful (kW) power delivered vs. the current drawn by the load, this is known as 'power factor'. The size of assets required is generally a function of the current. Therefore the closer the power factor is to 1.00, the smaller the asset required to supply the load.
ORG	Office of the Regulator General of Victoria
Rate shock	The change in average price to a customer caused by changes in price structure or level, compared to prices prior to the changes.
Revenue Yield	The form of the regulatory Distribution price control in effect until 31 December 2000 but now replaced by the ORG with the 'Tariff basket' approach. Distribution revenue was correlated to volume throughput on the Distribution system and tariff revenue was "corrected" to a regulated amount based on throughput from year to year via a "k-factor" mechanism.

SDIC **Summer Demand Incentive Charge**

Supply-side Similar to **demand-side** described earlier, but this time referring to **providers** of services in the chain supplying customers. In the case of this paper, the term refers specifically to electricity distributors and generally to the industry including transmission and generation.

Tariff Basket The form of the regulatory Distribution price control in effect since 1 January 2001. Distribution revenue is the same as tariff revenue. The regulated amount is the same as actual tariff revenue and the "correction factors" now apply only to transmission revenue.